

Cycle 19 Autonomous Report

Date: 2026-05-16

Repository: qc_exciton_LCC

Task selected

Add a scaling-summary mode to the synthesis table regeneration helper.

Major changes

Added a --summary mode that prints canonical log-log power-law fits for the regenerated synthesis intercept and workspace tables.

Added pure helper tests for the power-law fit and canonical summary formatter.

Files changed

scripts/regenerate_synthesis_tables.py

tests/test_regenerate_synthesis_tables.py

docs/cycle_reports/cycle19_report.pdf

Rendered Equations

This page is a rendering check for the inbox requirement that equations appear as LaTeX-style math.

$$I_1(M, N) \approx c_I(N)M^{\alpha_I}$$

$$W(M, N, b) \approx c_W(N, b)M^{\alpha_W}$$

$$\log y = \alpha \log M + \log c$$

Goals already achieved

Prior cycles already achieved analytic-vs-Bloq synthesis Toffoli and qubit-count guards over the tabulated grid.

Prior cycles already achieved reusable math-rendered autonomous PDF reports.

Verification

Tests and checks

PYTHONPATH=src python tests/test_regenerate_synthesis_tables.py -> 6/6 passed.

python -m py_compile scripts/regenerate_synthesis_tables.py
tests/test_regenerate_synthesis_tables.py -> passed.

PYTHONPATH=src python scripts/regenerate_synthesis_tables.py --no-workspace --summary ->
printed $I_1(n_{\text{blocks}}, N=256)$ power-law fit.

Achieved goal

Added a reportable empirical scaling diagnostic for synthesis table closed-form work.

Next recommended task

Use the new --summary output as a guide while deriving $I_1(n_{\text{blocks}}, n_{\text{rows}})$ from the QROAMClean optimizer expression.